

ALGEBRAIC FRACTIONS, (ADDING AND SUBTRACTING ALGEBRAIC

Name: _____

Date: _____

LO: I can add and subtract algebraic fractions by finding a common denominator.

What are algebraic fractions?

An **algebraic fraction** has a variable (letter) in the numerator, denominator, or both. To **add or subtract** them, we need a common denominator, just like with number fractions.

Strategy: Find the LCM of the denominators, convert each fraction, then add or subtract the numerators.

Quick examples

●	$\frac{1}{x} + \frac{2}{x} = \frac{3}{x}$	same denominator
●	$\frac{1}{x} + \frac{1}{y} = \frac{y + x}{xy}$	different denominators
●	$\frac{2}{x+1} + \frac{3}{x+2} = \frac{2(x+2) + 3(x+1)}{(x+1)(x+2)}$	linear denominators
●	$\frac{1}{x} + \frac{1}{y} = \frac{2}{x+y}$	cannot add denominators
●	$\frac{2}{x} + \frac{3}{x} = \frac{5}{x}$	denominator stays the

Key idea

Find the lowest common denominator (LCD), rewrite each fraction over the LCD, then combine the numerators.

$$\frac{a}{x} + \frac{b}{y} = \frac{ay + bx}{xy}$$

Cross-multiply to find common denominator xy

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - same denominator

Simplify $\frac{3}{x} + \frac{5}{x}$

Denominators are the same: x

Add numerators: $3 + 5 = 8$

Answer: $\frac{8}{x}$

$$\frac{8}{x}$$

When denominators match, simply add the numerators.

Example 2 - different single-term denominators

Simplify $\frac{2}{x} + \frac{3}{y}$

LCD = xy

Rewrite: $\frac{2y}{xy} + \frac{3x}{xy}$

Combine: $\frac{2y + 3x}{xy}$

$$\frac{2y + 3x}{xy}$$

Multiply each fraction by the other denominator over itself.

Example 3 - linear denominators

Simplify $\frac{1}{x+1} + \frac{2}{x+3}$

LCD = $(x+1)(x+3)$

Rewrite: $\frac{1(x+3) + 2(x+1)}{(x+1)(x+3)}$

Expand: $\frac{x+3 + 2x+2}{(x+1)(x+3)} =$

$\frac{3x + 5}{(x+1)(x+3)}$

$$\frac{3x + 5}{(x+1)(x+3)}$$

Expand the numerators carefully, then collect like terms.

Example 4 - subtraction

Simplify $\frac{5}{x+2} - \frac{3}{x+1}$

LCD = $(x+2)(x+1)$

Rewrite: $\frac{5(x+1) - 3(x+2)}{(x+2)(x+1)}$

Expand: $\frac{5x+5 - 3x-6}{(x+2)(x+1)} =$

$\frac{2x - 1}{(x+2)(x+1)}$

$$\frac{2x - 1}{(x+2)(x+1)}$$

With subtraction, distribute the minus sign carefully.

YOUR TURN – PRACTICE (deliberate practice)

1. Same denominator

Simplify each expression.

- a) $\frac{2}{x} + \frac{5}{x}$
- b) $\frac{7}{a} - \frac{3}{a}$
- c) $\frac{4}{x+1} + \frac{2}{x+1}$
- d) $\frac{9}{2x} - \frac{5}{2x}$

2. Different single-term denominators

Find a common denominator and simplify.

- a) $\frac{1}{x} + \frac{1}{y}$
- b) $\frac{3}{a} + \frac{2}{b}$
- c) $\frac{5}{x} - \frac{1}{2x}$
- d) $\frac{2}{3x} + \frac{1}{x}$

3. Linear expression denominators

Simplify. Write as a single fraction.

- a) $\frac{1}{x+1} + \frac{1}{x+2}$
- b) $\frac{2}{x+3} + \frac{1}{x+1}$
- c) $\frac{3}{x+4} - \frac{1}{x+2}$
- d) $\frac{4}{x-1} + \frac{2}{x+3}$

4. Harder subtraction

Simplify fully.

- a) $\frac{5}{x+2} - \frac{2}{x+1}$
- b) $\frac{3}{x-2} - \frac{1}{x+4}$
- c) $\frac{x}{x+1} + \frac{2}{x-1}$
- d) $\frac{4}{2x+1} - \frac{1}{x+3}$

5. Mixed algebraic fractions

Write as a single fraction in simplest form.

- a) $\frac{1}{x} + \frac{1}{x+1}$
- b) $\frac{3}{x+2} - \frac{2}{x-3}$
- c) $\frac{x}{x+1} - \frac{x}{x+2}$
- d) $\frac{2}{x-1} + \frac{3}{x+1}$
- e) $\frac{1}{x} + \frac{1}{x+1} + \frac{1}{x+2}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{1}{x} + \frac{2}{x} = \frac{3}{x}$ ☐

Correct answer: _____

Reason: _____

b) $\frac{1}{x} + \frac{1}{y} = \frac{2}{x+y}$ ☐

Correct answer: _____

Reason: _____

c) To add fractions with different denominators, find the LCM first ☐

Correct answer: _____

Reason: _____

d) $\frac{a}{b} + \frac{c}{d} = \frac{a+c}{b+d}$ ☐

Correct answer: _____

Reason: _____

e) $\frac{3}{x+1} - \frac{1}{x+1} = \frac{2}{x+1}$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Show that $\frac{1}{x-1} - \frac{1}{x+1}$ simplifies to $\frac{2}{x^2-1}$. Hence find the value when $x = 5$.

Expression: _____

Simplified: _____

b) Simplify $\frac{x+1}{x+2} + \frac{x+3}{x+4}$ and write your answer as a single fraction. What value(s) of x would make this expression undefined?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

